Assignment 12: Complete Production Deployment Project

Objective: Deploy a production-ready application environment implementing modern DevOps, monitoring, logging, backup, security, and operational best practices.

Project Overview

The project integrates a Dockerized application, MySQL database with persistent storage, Nginx reverse proxy, monitoring stack (Prometheus & Grafana), centralized logging (Loki & Promtail), automated backups, security hardening, and deployment automation practices suitable for production environments.

Repository Structure

assignment12-production-deployment/
■■■ app/
■■■ nginx/
■■■ mysql/
■■■ backups/
■■■ monitoring/
■ ■■■ prometheus/
■ ■■■ grafana/
■■■ logging/
■ ■■■ loki/
■ ■■■ promtail/
■■■ screenshots/
■■■ docker-compose.yml
■■■ deployment-guide.md
■■■ rollback-guide.md
■■■ troubleshooting-guide.md
■■■ README.md

Production Components Implemented

✓ Dockerized Application
✓ MySQL Database with Persistent Storage
✓ Nginx Reverse Proxy
✓ SSL/TLS Ready Configuration
✓ Monitoring using Prometheus & Grafana
✓ Centralized Logging using Loki & Promtail
✓ Automated Backup Strategy
✓ Linux Security Hardening
✓ Rollback and Recovery Procedures
✓ Production Deployment Architecture

Architecture Diagram (Logical)

Users → Nginx Reverse Proxy → Dockerized Application → MySQL Database
↓
Prometheus Monitoring
↓
Grafana
↓
Loki + Promtail Logging
↓
Automated Backups

Deployment Guide Summary

1. Build application image.
2. Configure Docker Compose services.
3. Start infrastructure components.
4. Configure monitoring and logging.
5. Validate application availability.
6. Verify backup procedures.
7. Perform production testing.

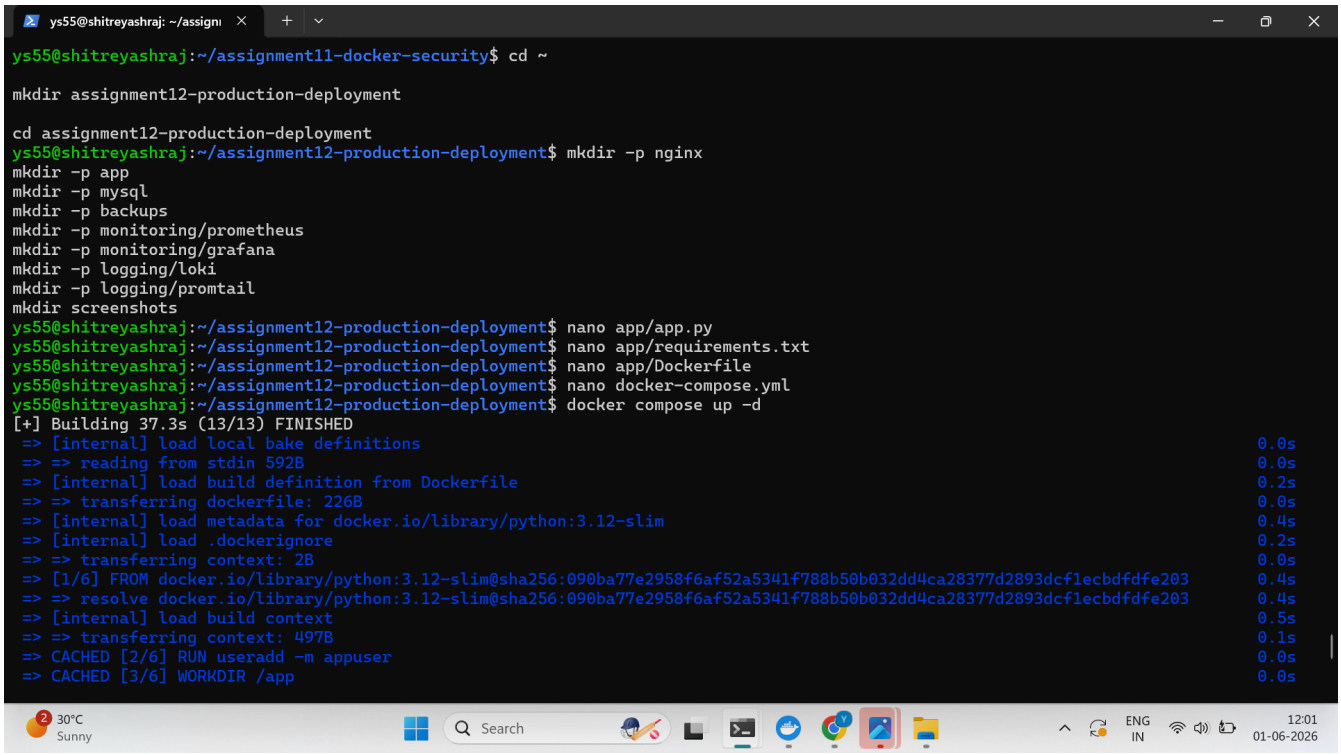
Rollback Guide Summary

- Restore previous Docker image version.
- Recover database from latest backup.
- Restart services using Docker Compose.
- Validate application functionality.
- Monitor logs and metrics after rollback.

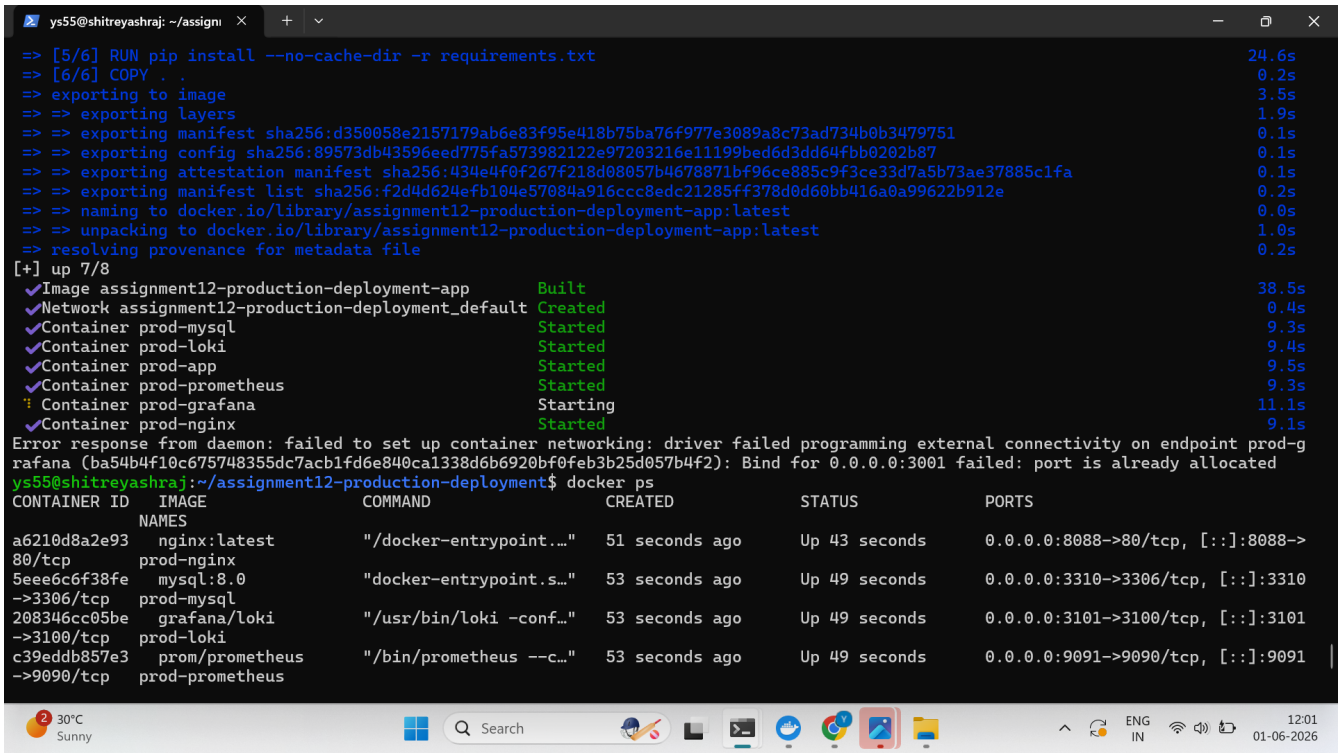
Troubleshooting Guide

- Port conflicts and container startup failures.
- Database connectivity issues.
- Monitoring service availability checks.
- Logging pipeline validation.
- Backup and restore verification steps.

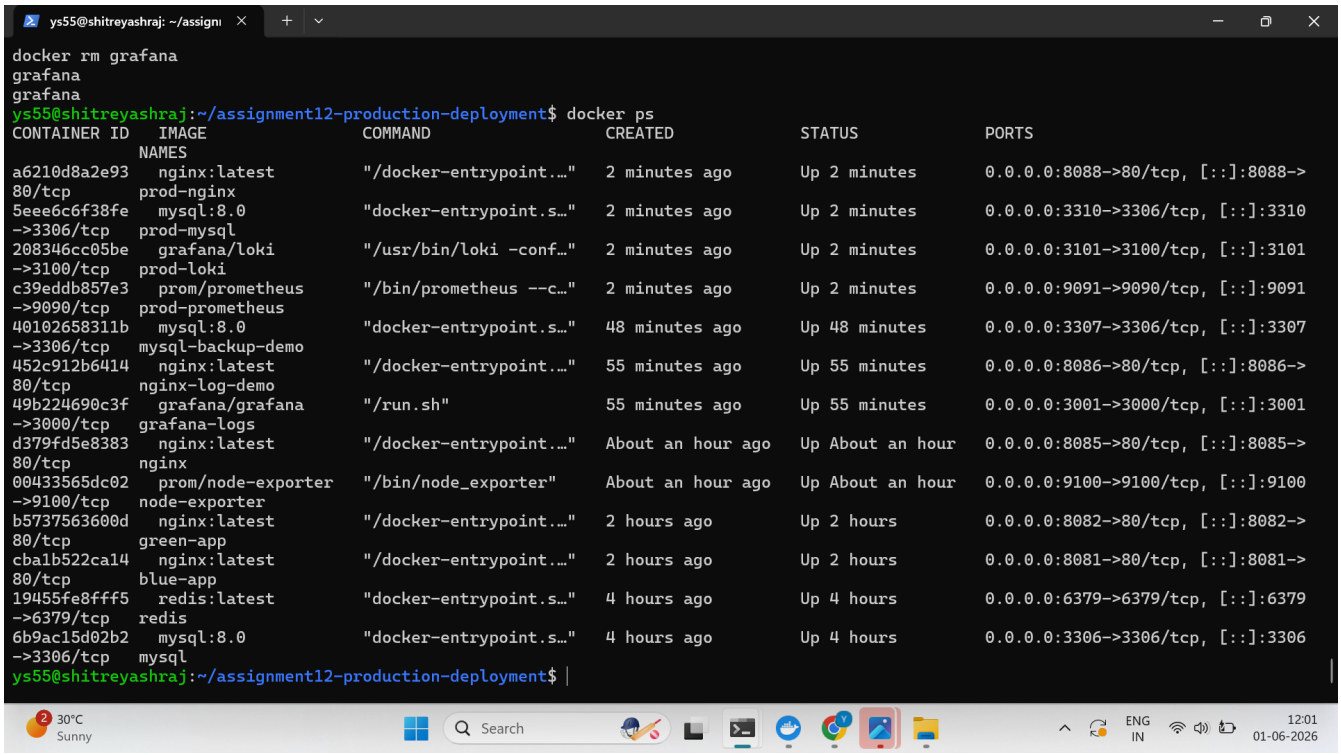
Implementation Evidence



Implementation Evidence



Implementation Evidence



Production Readiness Assessment

The environment demonstrates deployment automation, observability, security, backup management, and operational readiness. Monitoring, logging, database persistence, and infrastructure services were successfully integrated into a single deployment platform.

Conclusion

This assignment successfully delivers a complete production deployment environment. The implementation combines application hosting, monitoring, centralized logging, database management, backup automation, security controls, and operational documentation, providing a strong foundation for real-world production deployments.